Wednesday April 22

1. HW 11.6 due tonight. Questions?
2. Questions about power series worksheet 1?
3. Section 11.9
4. power series worksheet 2 (3 for HW)

$$\sum_{n=1}^{\infty} \frac{(x-3)^n}{n4^n} \qquad \lim_{n\to\infty} \frac{(x-3)^{n+1}}{(n+1)4^{n+1}} \cdot \frac{n4^n}{(x-3)^n}$$

$$= \lim_{n\to\infty} \frac{(x-3)^{\cancel{n+1}}}{\cancel{(x-3)^n}} \cdot \frac{\overset{1}{\cancel{n}}}{\cancel{n+1}} \cdot \frac{\cancel{4^n}}{4^{\cancel{n+1}}} = \frac{x-3}{4} \qquad -1 < \frac{x-3}{4} < 1$$

$$-4 < x-3 < 4$$

$$-1 < x < 7$$

$\underline{x=-1}$

$$\sum \frac{(-4)^n}{n4^n} = \sum \frac{(-1)^n(4)^n}{n \cdot 4^n} = \sum \frac{(-1)^n}{n} \boxed{C} \qquad \boxed{[-1, 7)}$$

$$\sum \frac{(7-3)^n}{n4^n} = \sum \frac{4^n}{n4^n} = \sum \frac{1}{n} \; D$$

$$\sum_{n=1}^{\infty} (n!)(x-4)^n$$

$$\lim_{n\to\infty} \frac{(n+1)!}{n!} \frac{(x-4)^{n+1}}{(x-4)^n} = \lim_{n\to\infty} \frac{(n+1)\cdot \cancel{n!}}{\cancel{n!}} \cdot \frac{\cancel{(x-4)^n}(x-4)}{\cancel{(x-4)^n}} = \begin{cases} \infty \\ -\infty \\ 0 \end{cases}$$

$$= \begin{cases} \infty & \text{if } x-4>0 \\ -\infty & \text{if } x-4<0 \\ 0 & \text{if } x-4=0 \end{cases} \Rightarrow \boxed{x=4}$$

11.6#3 $$\sum_{n=1}^{\infty} \frac{n+1}{2^{2n+4}} \qquad \lim_{n\to\infty} \frac{n+1+1}{2^{2(n+1)+4}} \cdot \frac{2^{2n+4}}{n+1}$$

$$= \lim_{n\to\infty} \frac{n+2}{n+1} \cdot \frac{2^{2n+4}}{2^{2n+6}} = \lim_{n\to\infty} \frac{\cancel{n}+\overset{1}{\cancel{2}}}{\cancel{n}+1} \cdot \frac{\cancel{2^{2n}} \cdot \cancel{2^4}}{\cancel{2^{2n}} \cdot 2^{\cancel{6}\,2}} = \frac{1}{4} < 1$$

$\therefore C$

11.9 Functions as Power Series

1 Functions as Power Series

Functions can be represented as power series.

- In this section we will look at certain functions that are relatively easy to represent as series.
- In section 11.10 we will look at representing more general functions as series.

Recall that the geometric series with 1st term a and common ratio r is

$$\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \cdots = \frac{a}{1-r}$$

The functions in the following examples are all similar to $\frac{a}{1-r}$, and we can use that to write them as a power series.

1.0.1 Example:

$$\frac{1}{1-2x} = \overset{0}{1} + \overset{1}{2x} + \overset{2}{4x^2} + \overset{3}{8x^3} + \overset{4}{16x^4} + \cdots = \sum_{n=0}^{\infty} \overset{a}{1}\,(\overset{r}{2x})^n$$

$a = 1$
$r = 2x$

$$\frac{a}{1-r}$$

1.0.2 Example: $a = 3 \quad r = -x^2$

$$\frac{3}{1+x^2} = \frac{3}{1-(-x^2)} = 3 - 3x^2 + 3x^4 - 3x^6 + 3x^8 - \cdots$$

$$= \sum_{n=0}^{\infty} 3(-x^2)^n = \sum_{n=0}^{\infty} 3(-1)^n (x^2)^n = \boxed{\sum_{n=0}^{\infty} (-1)^n \cdot 3 \cdot x^{2n}}$$

$$a = x^3 \quad r = 2x$$

1.0.3 Example:

$$\frac{x^3}{1-2x} = \overset{0}{x^3} + \overset{1}{2x^4} + \overset{2}{4x^5} + \overset{3}{8x^6} + \overset{4}{16x^7} + \cdots$$

$$= \sum_{n=0}^{\infty} x^3 (2x)^n = \boxed{\sum_{n=0}^{\infty} 2^n x^{n+3}}$$

$$x^3 \cdot 2^n \cdot x^n$$

1.0.4 Example:

$$a = \tfrac{1}{4}x \quad r = \tfrac{1}{4}x$$

$$\frac{x}{4-x} \cdot \frac{\frac{1}{4}}{\frac{1}{4}} = \frac{\frac{1}{4}x}{1 - \frac{1}{4}x} = \overset{0}{\frac{1}{4}x} + \overset{1}{\frac{1}{16}x^2} + \overset{2}{\frac{1}{64}x^3} + \overset{3}{\frac{1}{256}x^4} + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{1}{4}x\left(\frac{1}{4}x\right)^n = \sum_{n=0}^{\infty} \frac{1}{4}x \cdot \frac{x^n}{4^n} = \boxed{\sum_{n=0}^{\infty} \frac{x^{n+1}}{4^{n+1}}}$$

1.0.5 Example:

$$\frac{3x^2}{x^2+5} = \frac{3x^2}{5+x^2} \cdot \frac{1/5}{1/5} = \frac{\frac{3}{5}x^2}{1+\frac{1}{5}x^2} = \frac{\frac{3}{5}x^2}{1-\left(-\frac{1}{5}x^2\right)} \qquad a = \frac{3}{5}x^2 \quad r = -\frac{1}{5}x^2$$

$$= \overset{0}{\frac{3}{5}x^2} - \overset{1}{\frac{3}{25}x^4} + \overset{2}{\frac{3}{125}x^6} + \cdots$$

$$= \sum_{n=0}^{\infty} \left(\frac{3}{5}x^2\right)\left(-\frac{1}{5}x^2\right)^n = \sum_{n=0}^{\infty} (-1)^n \frac{3x^2 x^{2n}}{5 \cdot 5^n} = \boxed{\sum_{n=0}^{\infty} (-1)^n \frac{3x^{2n+2}}{5^{n+1}}}$$

2 Derivatives and Integrals with Power Series

Goal: Find a power series for $\dfrac{1}{(1-x)^2}$

First, observe that

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) \quad = \quad \frac{d}{dx}\left((1-x)^{-1}\right) \quad = \quad -(1-x)^{-2}(-1) \quad = \quad \frac{1}{(1-x)^2}$$

Plan: Get the power series for $\dfrac{1}{1-x}$, then take its derivative.

2.0.1 Example:

Find the power series representation for $\arctan(x)$, and find its interval of convergence.

$$\left(\text{Hint: } \frac{d}{dx}\left(\arctan(x)\right) = \frac{1}{1+x^2}\right)$$